

Westhaven Cancer Institute

Breast Pathology and Molecular Diagnostics
4500 Cypress Boulevard, Westhaven, IL 60004 • CLIA 14D9988440

BREAST SURGICAL PATHOLOGY — PARTIAL MASTECTOMY WITH SENTINEL NODE BIOPSY

Patient: BERGSTROM, ANITA L.	Accession: WCI-26-S-04115
MRN: MRN-5519082	Collected: 2026-04-22
DOB / Age / Sex: 1973-09-30 / 52 y / Female	Reported: 2026-04-29
Surgeon: Okafor, Tasha M., MD (Breast Surgical Oncology)	Location: Outpatient — Breast Cancer Center

FINAL SURGICAL PATHOLOGY DIAGNOSIS

A. Left breast, partial mastectomy:

- **Invasive carcinoma, no special type (ductal); 2.1 cm.**
- **Nottingham grade 2 (tubule 3 + nuclear pleomorphism 2 + mitoses 2 = 7/9).**
- **No lymphovascular invasion identified.**
- **Margins: NEGATIVE; closest 4 mm (anterior).**
- **Associated ductal carcinoma in situ, intermediate grade, solid pattern, occupying ~10% of the tumor.**

B. Sentinel lymph nodes (3, level I + level II + internal mammary):

- **NEGATIVE for metastatic carcinoma (0/3, by H&E and pankeratin IHC).**

CLINICAL HISTORY

52-year-old postmenopausal woman, gravida 2 para 2. Screening mammography (annual) demonstrated a new 2.1 cm spiculated mass in the upper outer quadrant of the left breast (BI-RADS 5). Core needle biopsy showed invasive carcinoma. No personal history of breast or ovarian cancer. Mother diagnosed with breast cancer at age 68 (receptor status unknown). Maternal aunt with ovarian cancer at 62. Patient elected breast-conserving surgery with sentinel lymph node biopsy on 2026-04-22.

GROSS DESCRIPTION

A. The specimen is labeled "left breast, lumpectomy." It consists of fibrofatty breast tissue measuring 7.5 x 6.0 x 3.1 cm. The specimen is oriented with sutures (short = superior, long = lateral). Serial sectioning reveals a firm, spiculated, gray-white tumor measuring 2.1 x 1.9 x 1.6 cm located in the upper outer aspect, approximately 4 mm from the anterior margin. All margins are inked appropriately. Representative sections are submitted in cassettes A1-A14.

B. Three sentinel lymph nodes labeled "level I," "level II," and "internal mammary," measuring 1.4 cm, 1.0 cm, and 0.8 cm respectively. Each bisected and submitted entirely (B1-B3).

MICROSCOPIC DESCRIPTION

Sections of the breast lesion (A1-A8) demonstrate an invasive carcinoma composed of nests and cords of moderately atypical epithelial cells infiltrating fibrous stroma. The architectural pattern is predominantly tubular and trabecular without well-formed glandular structures in >75% of the tumor (tubule formation score 3). Nuclei show moderate pleomorphism with vesicular chromatin and small nucleoli (nuclear score 2). Mitoses are counted at 8 per 10 high-power fields (40x, 0.50 mm field diameter; mitotic score 2). Combined Nottingham score is 3 + 2 + 2 = 7 of 9, corresponding to histologic grade 2. No definite lymphovascular invasion is identified after review of all sections including peri-tumoral D2-40 IHC. Associated ductal carcinoma in situ, intermediate grade, solid and cribriform pattern, is present and accounts for approximately 10% of the tumor; necrosis is not present in the in-situ component. All inked margins are free of tumor; the closest margin is 4 mm anteriorly. Sections of the three sentinel lymph nodes (B1-B3) are entirely submitted and show no metastatic carcinoma by H&E or by pankeratin (AE1/AE3) IHC.

BIOMARKER STUDIES (IHC)

Marker	Result	Score / Quantitation	Status
Estrogen receptor (ER, SP1)	Positive, 95% strong nuclear	Allred 8/8 (proportion 5 + intensity 3)	POSITIVE
Progesterone receptor (PR, 1E2)	Positive, 40% strong nuclear	Allred 7/8 (proportion 4 + intensity 3)	POSITIVE
HER2 (4B5)	1+ membranous in <10% of cells	Incomplete, faint; circumferential <10%	NEGATIVE (no FISH indicated)

Ki-67 (MIB-1)	~18% in hot-spot, ~12% overall	Intermediate proliferation index	Reported per ASCO/CAP 2020 guidance
E-cadherin	Retained, membranous	Excludes invasive lobular carcinoma	Retained
AE1/AE3 (sentinel nodes)	Negative	No metastatic carcinoma identified	Negative

HER2 IHC 1+ is reported as *NEGATIVE* per ASCO/CAP 2018; no reflex FISH is indicated. The Ki-67 result is provided per ASCO/CAP 2020 guidance but is not used to assign a definitive cutoff for endocrine-only vs combined therapy decisions.

SYNOPTIC SUMMARY (CAP-style, key elements)

Procedure	Partial mastectomy with sentinel lymph node biopsy
Tumor site	Left breast, upper outer quadrant
Tumor size (largest invasive focus)	2.1 cm
Histologic type	Invasive carcinoma, no special type (ductal)
Nottingham score (tubule + nuclear + mitoses)	3 + 2 + 2 = 7 / 9 → Grade 2
Lymphovascular invasion	Not identified
DCIS component	Present, intermediate grade, ~10%
Margin status	Negative; closest 4 mm anteriorly
Lymph node status	0 / 3 sentinel nodes positive
Pathologic stage (AJCC 8th)	pT2 pN0 (no distant assessment)

COMMENT

Invasive ductal carcinoma, no special type, Nottingham grade 2, 2.1 cm, ER+/PR+/HER2-negative, with associated intermediate-grade DCIS (~10%). Margins are negative (closest 4 mm anterior) and 0/3 sentinel nodes are positive. Pathologic stage is pT2 pN0. Companion molecular profiling (PIK3CA hotspot, ERBB2 copy number) and a 70-gene recurrence-risk assay have been ordered (accession WCI-26-MOL-08812 and GRRS-26-RR-00731 respectively). Outside germline panel (BRCA1/2/PALB2/CHEK2/ATM/TP53) ordered per family history; result pending (accession ATLAS-26-G-06224). Integrated diagnosis to follow once all ancillary studies are available.

Electronically signed: Okafor, Tasha M., MD — Breast Pathology. Co-reviewed: Liu, Hong-Mei, MD, PhD (Molecular Pathology Director). Report status: FINAL (surgical pathology component).

Page 1 of 1 • Accession WCI-26-S-04115 • MRN-5519082